

Vitamins and Minerals

Vitamin D (GC, CYP2R1, DHCR7, CYP24A1, VDR), B12 and folate (FUT2, TCN2, MTHFR), vitamin A, E, K, C and B6, choline, long chain omega-3, calcium and phosphate

Sérgio F · 02-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

Contents

Section	Page
1. Summary	3
2. How to read this report	4
3. Polygenic scores	5
4. Key variants by topic	6
5. GWAS loci burden	8
6. Favourable	9
7. Unfavourable	10
8. Mechanisms of ageing	11
9. Practical suggestions	12
10. Limitations	13
11. Methods and sources	14
12. Annex: all variants analysed	15

1. Summary

Favourable. Genetic vitamin D is high: three scores in the percentiles 84.7, 84.0 and 91.2, with GC and DHCR7 without low-level alleles. The conversion of vegetable omega-3 to EPA/DHA is efficient (FADS1 and ELOVL2 G/G). It does not have the variant SVCT1 of low absorption of vitamin C, nor the BCO1 A379V, it has favourable CUBN and MTRR, and the load of folate loci is high (96.5).

Unfavourable. MTHFR is compound heterozygous (C677T + A1298C; see §13), TCN2 P259R is heterozygous (slightly worse B12 cell transport) and the B12 score is at percentile 34.0. BCO1 R267S is heterozygous (less conversion of beta-carotene into vitamin A). **ALPL C/T** slightly lowers the active B6. PEMT V175M is heterozygous (less endogenous choline). CYP2R1 and CYP24A1 are heterozygous, but do not weigh against the high scores.

Pharmacogenetics. VKORC1 –1639 T/T (Imputed, DR2 0.99): high sensitivity to warfarin, which would require a lower dose ~50%. Direct anticoagulants do not depend on this gene. The doctor must know.

What does it mean? The micronutrient genes are globally good. High genetic vitamin D, added to D3 10 000 IU/day, advises to monitor 25(OH)D and calcium to not exceed the useful interval. The points to be covered by the diet and plan are folate and active B12 (MTHFR + TCN2), choline (already in S38) and vitamin A from foods (without supplements of retinol, by interaction with isotretinoin).

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Key variants** 22 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 21 of the chip, 1 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 3 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 7 PGS Catalog scores calculated locally with 4,544,330 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 3887 loci, 3185 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In vitamin D and B12, a high percentile = higher genetic level (favourable). For calcium and phosphate there is no judgment (the organism regulates them tightly).

Vitamin D

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
25(OH)D (vitamin D)	PGS000882 (Revez 2020, SbayesR, 1.09 million variants)	1,069,187 1,094,650 (weighted coverage 97%)	84.7	HIGH	favourable
Vitamin D	PGS002721 (Weissbrod 2022, SbayesR, 982 thousand variants)	960,019 981,998 (weighted coverage 97%)	84.0	HIGH	favourable
25(OH)D (vitamin D)	PGS019960 (Chen X 2024, 62 variants)	54 62 (weighted coverage 88%)	91.2	HIGH	favourable

Vitamin B12

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Vitamin B12	PGS003158 (Ma 2022, PRS-CS, 1.11 million variants)	1,082,449 1,113,819 (weighted coverage 96%)	34.0	TYPICAL	typical

Calcium and phosphate

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Serum calcium	PGS001893 (Privé 2022, 48 thousand variants)	46,656 48,395 (weighted coverage 96%)	22.3	TYPICAL	without judgment
Serum calcium	PGS002106 (Privé 2022, LDpred2, 702 thousand variants)	689,671 702,009 (weighted coverage 97%)	8.7	LOW	without judgment
Serum phosphate	PGS002216 (Privé 2022, LDpred2, 709 thousand variants)	696,294 708,693 (weighted coverage 98%)	53.1	TYPICAL	without judgment

4. Key variants by topic

Variants with replicated effect on vitamin and mineral levels and on the pharmacogenetics of vitamin K.

Vitamin D

The level of 25(OH)D depends on the transport (GC), the synthesis in the skin (DHCR7), liver activation (CYP2R1) and degradation (CYP24A1). VDR is the receiver.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
GC (vitamin D binding protein)	rs2282679		T/T	chip	G (0 copy/s)	favourable	The G allele lowers the binding protein and the circulating 25(OH)D (~-5 nmol/L by allele)	Strong
CYP2R1 (25-hydroxylase)	rs10741657		A/G	chip	G (1 copy/s)	unfavourable (light)	G allele reduces hepatic 25-hydroxylation (25(OH)D ↓)	Strong
DHCR7	rs12785878		T/T	chip	G (0 copy/s)	favourable	The G allele maintains more 7-desidrocholesterol to be converted into cholesterol, with less vitamin D produced in the skin	Moderate
CYP24A1 (24-hydroxylase)	rs6013897		T/A	chip	A (1 copy/s)	unfavourable (light)	Vitamin D catabolism: A allele is associated with lower 25(OH)D	Moderate
VDR	rs2228570	FokI	A/G	chip	A (1 copy/s)	without judgment	FokI (allele A = f): longer and less active vitamin D receptor (small effect, mixed replication)	Weak

Vitamin B12 and folate

The absorption and transport of B12 (FUT2, CUBN, TCN2) and the cycle of folate and methionine (MTHFR, MTRR). The methylation in detail is in §13.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
FUT2	rs601338	W154X	G/A	chip	A (1 copy/s)	neutral	W154X: Allele A (non-secretor in A/A) is associated with higher B12 (less H. pylori and other flora)	Strong
TCN2	rs1801198	P259R	G/C	chip	G (1 copy/s)	unfavourable (light)	P259R: the allele G low to holotranscobalamine (active B12 in cells)	Moderate
CUBN	rs1801222		G/G	chip	A (0 copy/s)	favourable	Intestinal receptor of the intrinsic B12-factor complex: allele A is associated with lower B12	Moderate
MTHFR	rs1801133	C677T	G/A	chip	A (1 copy/s)	unfavourable (light)	C677T: less 5-MTHF, erythrocyte folate ↓ and homocysteine ↑ (methylation pathway detail)	Strong
MTHFR	rs1801131	A1298C	T/G	chip	G (1 copy/s)	unfavourable (light)	A1298C: activity of MTHFR slightly ↓ (relevant combined with C677T)	Moderate
MTRR	rs1801394	I22M	A/A	chip	G (0 copy/s)	favourable	I22M: regeneration of methionine synthase dependent on B12	Weak

Liposoluble vitamins A, E and K

The conversion of beta-carotene into vitamin A (BCO1), the metabolism of vitamins E and K (CYP4F2) and the target of vitamin K (VKORC1).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
BCO1	rs12934922	R267S	A/T	chip	T (1 copy/s)	unfavourable (light)	R267S: lower conversion of beta-carotene to vitamin A	Moderate
BCO1	rs7501331	A379V	C/C	chip	T (0 copy/s)	favourable	A379V: lower beta-carotene conversion (additive effect to R267S)	Moderate
CYP4F2	rs2108622	V433M	C/C	chip	T (0 copy/s)	without judgment	V433M: less omega-hydroxylation of vitamins E and K (light tocopherol and vitamin K ↑; warfarin: dose ↑)	Moderate
VKORC1	rs9923231	–1639G>A	T/T	imputation DR2=0.99	T (2 copy/s)	without judgment	–1639G>A: less vitamin K epoxide reductase (sensitivity to warfarin ↑)	Strong
APOA5 (region)	rs12272004		C/C	chip	A (0 copy/s)	neutral	Allele A is associated with higher alpha-tocopherol (lipid transport)	Moderate

Vitamin C, B6 and choline

The intestinal transporter of vitamin C (SVCT1), alkaline phosphatase that degrades the active form of B6 (ALPL) and endogenous choline synthesis (PEMT).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
SLC23A1 (SVCT1)	rs33972313	V264M	C/C	chip	T (0 copy/s)	favourable	V264M: less renal vitamin C resorption (plasma ascorbate ↓); rare	Moderate
ALPL	rs4654748		C/T	chip	C (1 copy/s)	unfavourable (light)	Alkaline phosphatase: C allele is associated with lower circulating vitamin B6 (PLP)	Moderate
PEMT	rs7946	V175M	C/T	chip	T (1 copy/s)	unfavourable (light)	V175M: less endogenous synthesis of phosphatidylcholine (more need for choline)	Moderate

Omega-3 long chain

The desaturases and elongases (FADS1, ELOVL2) convert the vegetable omega-3 into EPA and DHA.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
FADS1	rs174537		G/G	chip	T (0 copy/s)	favourable	The T allele reduces the activity of FADS1 (less EPA/DHA endogenous from ALA)	Strong
ELOVL2	rs953413		G/G	chip	A (0 copy/s)	favourable	Omega-3 elongation: allele A is associated with lower DHA	Moderate

Chalk

The calcium sensor (CASR) defines the point of regulation of calcium in the blood. Iron, zinc, copper, selenium and magnesium are in §33.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CASR	rs1801725	A986S	G/G	chip	T (0 copy/s)	without judgment	A986S: the T allele lowers the sensitivity of the calcium sensor (Slightly ↑ serum calcium)	Moderate

5. GWAS loci burden

Sum of all loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of each level, with alleles oriented towards higher value, compared to the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Vitamin D (25(OH)D)	461 (374)	411 of 748	373	80.4	favourable
Vitamin B12	233 (157)	159 of 314	157	67.1	typical
Folate	23 (21)	17 of 42	21	96.5	favourable
Homocysteine	40 (37)	33 of 74	32	67.5	typical
Retinol and carotenoids	69 (63)	57 of 126	63	32.9	typical
Vitamin E	25 (18)	14 of 36	18	54.3	without judgment
Omega 3 (EPA/DHA)	2230 (1817)	1004 of 3634	1815	73.1	typical
Serum calcium	716 (617)	596 of 1234	615	28.8	without judgment
Serum magnesium	90 (81)	61 of 162	80	2.1	without judgment

6. Favourable

- **High genetic vitamin D:** percentiles 84.7, 84.0 and 91.2; loci load at 80.4. **GC rs2282679 T/T** (normal binding protein) and **DHCR7 T/T** (normal skin synthesis). **Strong.**
- **Omega 3:** FADS1 rs174537 G/G and ELOVL2 G/G, the most efficient conversion of ALA into EPA/DHA; loci load at 73.1. **Moderate.**
- **Vitamin C:** SLC23A1 V264M absent (C/C). **Moderate.**
- **B12:** CUBN G/G (without low absorption allele); FUT2 secretor (G/A; neutral). **MTRR I22M A/A:** without the methionine synthase reductase risk allele. **Weak moderate.**
- **Folate:** folate loci load at percentile 96.5 (low loci; partially compensates for MTHFR). **Weak moderate.**
- **BCO1 A379V absent.** **Weak.**

7. Unfavourable

- **MTHFR C677T G/A + A1298C T/G** (compound heterozygous): activity of MTHFR ~50-60%. Homocysteine only slightly ↑ if folate or riboflavin are low; loci load of homocysteine at 67.5. Detail in §13. **Moderate.**
- **TCN2 P259R G/C**: less B12 bound to transcobalamin (active B12) for each C allele. B12 score at percentile 34.0; loci load at percentile 67.1. **Weak moderate.**
- **BCO1 R267S A/T**: conversion of beta-carotene to ~30% lower by risk allele. Relevant because it cannot take retinol. **Weak moderate.**
- **PEMT V175M C/T**: less choline synthesis in the liver, then more dependence on the diet choline and supplements. **Weak moderate.**
- **ALPL rs4654748 C/T**: B6 active (PLP) slightly lower. **Weak.**
- **CYP2R1 A/G and CYP24A1 T/A**: 1 lower vitamin D allele in each, compensated by the high scores. **Weak.**
- **Low serum calcium** (percentiles 22.3 and 8.7; loci load at 28.8) and low serum magnesium (loci load at 2.1): without judgment, within normal regulation; magnesium is in §33. **Information.**

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Osteoporosis	High genetic vitamin D (favourable); low genetic serum calcium, without judgment	Favourable (light)
Immunosenescence	High vitamin D supports innate and adaptive immunity	Favourable (light)
Epigenetic Changes	MTHFR compound heterozygous and TCN2: less methyl donors if folate or B12 are missing (see §13)	Unfavourable (mild)
Vascular Aging	Homocysteine (MTHFR/TCN2) against a good conversion of omega-3 (FADS1/ELOVL2)	Mixed
Chronic Inflammation	FADS1/ELOVL2 efficient (plus EPA/DHA); high vitamin D	Favourable (light)
Oxidative stress	Normal SVCT1 (vitamin C); BCO1 R267S (less vitamin A from carotenoids)	Mixed
Metabolic Homeostasis Decline	PEMT V175M: more need for choline to export liver fat	Unfavourable (mild)

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Vitamin D3 10 000 IU (current plan):** with high-level genetics, measuring 25(OH)D and serum calcium 2×/year. The objective is to stay in the useful interval without exceeding it; if 25(OH)D passes the upper limit, talk to the doctor about the dose. Maintain vitamin K2 if already in the plan. *Suggestion (not applied).*
2. **Folate and good quality B12 (MTHFR + TCN2):** high dietary folate (green leaf vegetables, legumes; the more fiber, the better). If using B supplement, prefer the methylfolate and methyl/adenosylcobalamine forms, with sufficient riboflavin (B2). Measure homocysteine and active B12 (holotranscobalamine or methylmalonic acid) once. See §13. *Suggestion (not applied).*
3. **Hill: keeping the S38 of the plan** (citicoline), consistent with PEMT C/T. Eggs and legumes also help. *Suggestion (not applied).*
4. **Vitamin A only from food** (without retinol supplements, by interaction with isotretinoin). With BCO1 R267S, betacarotene conversion is lower. Carotenoids of the plan (lutein 20 mg + zeaxanthin 4 mg, astaxanthin 12 mg) are not pro-vitamin A and remain. *Suggestion (not applied).*
5. **B6:** food sources (leguminous, fish, potato). No megadoses (neuropathy). *Suggestion (not applied).*
6. **Omega-3 3.2-4 g/day (already in the plan):** with efficient FADS1/ELOVL2, the Omega-3 index must respond well. Measure it once to confirm. *Suggestion (not applied).*
7. **VKORC1 T/T:** register in the clinical process the high sensitivity to warfarin. If you ever need anticoagulation, the doctor may prefer a direct anticoagulant or adjust the dose of warfarin by the genotype. Keep the vitamin K intake stable. *Suggestion (not applied).*
8. **Zinc 20-25 mg/day total with copper (already in the plan).** Iron, zinc, copper, selenium and magnesium are in §33; ferritin below 100. *Suggestion (not applied).*

10. Limitations

- The scores and variants explain only a part of the levels. The diet, supplementation and sun weigh more, and the measured value orders.
- VKORC1 –1639 was imputed (DR2 0.99). Before a dose decision of warfarin, confirm the genotype in a clinical laboratory.
- The load of magnesium and calcium loci includes renal and parathyroid regulation loci. It is descriptive.
- The chip does not read rare variants (e.g. CUBN, TCN2 or CYP2R1 of loss of function, or hypophosphatesia ALPL).
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The published allele of effect is compared with the alleles of the imputed VCF; in the palindromic SNPs the strand + (standard of the GWAS Catalog and the PGS Catalog) is assumed.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog (7 scores: vitamin D, B12, calcium and phosphate). Score = Δ coefficient $\times$ dosage. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog ($p \leq 5 \times 10^{-8}$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** reference literature (Wang 2010 and Revez 2020 for vitamin D; Hazra 2008 for FUT2 and B12; Tanaka 2009 for B6, B12 and folate; Leung 2009 for BCO1; CPIC 2017 for VKORC1 and CYP4F2), GWAS Catalog and ClinVar. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Vitamin D	GC (vitamin D binding protein)	rs2282679	T/T	chip	G	0	strand +	without the risk allele
Vitamin D	CYP2R1 (25-hydroxylase)	rs10741657	A/G	chip	G	1	strand +	1 copy of risk allele
Vitamin D	DHCR7	rs12785878	T/T	chip	G	0	strand +	without the risk allele
Vitamin D	CYP24A1 (24-hydroxylase)	rs6013897	T/A	chip	A	1	strand + (palindromic; assumed +)	1 copy of risk allele
Vitamin D	VDR	rs2228570	A/G	chip	A	1	strand +	1 copy of effect allele
Vitamin B12 and folate	FUT2	rs601338	G/A	chip	A	1	strand +	G/A: secretor (the effect on B12 is recessive)
Vitamin B12 and folate	TCN2	rs1801198	G/C	chip	G	1	strand + (palindromic; assumed +)	1 copy of risk allele
Vitamin B12 and folate	CUBN	rs1801222	G/G	chip	A	0	strand +	without the risk allele
Vitamin B12 and folate	MTHFR	rs1801133	G/A	chip	A	1	strand +	1 copy of risk allele
Vitamin B12 and folate	MTHFR	rs1801131	T/G	chip	G	1	strand +	1 copy of risk allele
Vitamin B12 and folate	MTRR	rs1801394	A/A	chip	G	0	strand +	without the risk allele
Liposoluble vitamins A, E and K	BCO1	rs12934922	A/T	chip	T	1	strand + (palindromic; assumed +)	1 copy of risk allele
Liposoluble vitamins A, E and K	BCO1	rs7501331	C/C	chip	T	0	strand +	without the risk allele
Liposoluble vitamins A, E and K	CYP4F2	rs2108622	C/C	chip	T	0	strand +	0 copy of the effect allele
Liposoluble vitamins A, E and K	VKORC1	rs9923231	T/T	imputation DR2=0.99	T	2	strand +	2 copy of the effect allele
Liposoluble vitamins A, E and K	APOA5 (region)	rs12272004	C/C	chip	A	0	strand +	without the protective allele (common genotype)
Vitamin C, B6 and choline	SLC23A1 (SVCT1)	rs33972313	C/C	chip	T	0	strand +	without the risk allele
Vitamin C, B6 and choline	ALPL	rs4654748	C/T	chip	C	1	strand +	1 copy of risk allele
Vitamin C, B6 and choline	PEMT	rs7946	C/T	chip	T	1	strand +	1 copy of risk allele
Omega-3 long chain	FADS1	rs174537	G/G	chip	T	0	strand +	without the risk allele
Omega-3 long chain	ELOVL2	rs953413	G/G	chip	A	0	strand +	without the risk allele
Chalk	CASR	rs1801725	G/G	chip	T	0	strand +	0 copy of the effect allele